Auxiliary antitumor effects of fungal proteins from *Hericium erinaceus* by target on the gut microbiota

Dongdong Wang, Xiangxiang Zhu, Xiaocui Tang, Hongye Li, Xie Yizhen, and Diling Chen

Abstract: Cancer represents a major disease burden worldwide. Despite continuous advances obtained in medical therapies recently, resistance to standard drugs and adverse effects still represent important causes of therapeutic failure. There is growing evidence that the gut microbiota can affect the response to chemo- and immunotherapeutic drugs by modulating efficacy and/or toxicity, and diet is the most important factor affecting the gut microbiota. In this study, we assessed the auxiliary antitumor effects of immunomodulatory fungal proteins from *Hericium erinaceus* (HEP) administered with the chemotherapy drug 5-Fluorouracil (5-Fu), and we attempted to identify new potential prebiotic bacteria for auxiliary antitumor treatment. There were 1,455 proteins identified from *H. erinaceus*. In a xenografted mouse model of cancer, HEP with 5-Fu significantly suppressed tumor growth, inhibited inflammatory markers such as interferon (IFN)-γ, interleukin (IL)-1β, IL-2, IL-6, tumor necrosis factor (TNF)-α, and lipopolysaccharide (LPS), and regulated the expression of Akt, CCDN1, CKD4, FOXM1, MMP7, MYC, PPAR-α, and PPAR-γ. 16S rRNA sequencing showed that HEP ameliorated the dysbacteriosis induced by 5-Fu, as it inhibited certain aerobic and microaerobic bacteria including *Parabacteroides*, *Flavobacteriaceae*, *Christensenellaceae*, *Anoxybacillus*, *Aggregatibacter*, *Comamonadaceae*, *Planococcaceae*, *Desulfovibrionaceae*, *Sporosarcina*, *Staphylococcus*, *Aerococcaceae*, and *Bilophila* in the xenografted mice, and increase some probiotic bacteria such as *Bifidobacterium*, *Gemellales*, *Blautia*, *Sutterella*, *Anaerostipes*, *Roseburia*, *Lachnobacterium*, *Lactobacillus*, and *Desulfovibrio*. This demonstrates that HEP could promote the antitumor efficacy of 5-Fu by improving the microbiota composition, the immune inflammatory response, and homeostasis.

Keywords: auxiliary antitumor effects, chemotherapy response, fungal immunomodulatory protein, gut microbiota, immunotherapeutic drugs

1. INTRODUCTION

Colorectal cancer is one of the most common malignant tumors. Most colorectal cancers are due to old age and lifestyle factors, with only a small number of cases being due to underlying genetic disorders. Risk factors include diet, obesity, smoking, and lack of physical activity (Azeem et al., 2015; Crosbie et al., 2018; Wang et al., 2018). Dietary factors that increase the risk include red and processed meat as well as alcohol (Rossi, Mirbagheri, Keshavarzian, & Bishehsari, 2018). Another risk factor is inflammatory bowel disease, which includes Crohn's disease and ulcerative colitis (Clarke & Feuerstein, 2018; Stidham & Higgins, 2018). Inherited genetic disorders that can cause colorectal cancer include familial adenomatous polyposis and hereditary nonpolyposis colon cancer; however, these represent <5% of cases (Samadder, Jasperson, & Burt, 2015). Colorectal cancer typically starts as a benign tumor, often in the form of a polyp, which over time becomes cancerous.

There is increasing consensus regarding the important role of the gut microbiota in cancer occurrence and development. Carcinogenic bacteria, metabolic products of intestinal microorganisms, and chronic inflammatory reactions have been identified as factors that promote the occurrence and development of cancer (Ericsson et al., 2015; Jahani-Sherafat, Alebouyeh, Moghim, Ahmadi Amoli, & Ghasemian-Safaei, 2018; Meng, Bai, Brown, Hood, & Tian, 2018). In the 1990s, scientists discovered that *Helicobacter pylori* was associated with gastric cancer. In 1994, *H. pylori* was classified as a Class I carcinogen, and research also showed that eradicating *H. pylori* could successfully prevent the occurrence of gastric cancer (Vohlonen et al., 2016). In addition, research confirmed that *Streptococcus*, *Enterococcus*, *Bacteroides fragilis*, *Fusobacterium*, and *Helicobacter hepaticus* were associated with colorectal, breast, prostate, and liver cancer (Biragyn & Ferrucci, 2018). Certain strains of the *Streptococcus bovis/equinus* complex are consumed by millions of people daily and thus may be safe (Marmolin et al., 2016). However, 25% to 80% of people with *S. bovis/gallolyticus* bacteremia have concomitant colorectal tumors, and *S. bovis/gallolyticus* seropositivity is a candidate practical marker for the early prediction of an underlying bowel lesion in high-risk populations (Chirouze et al., 2013).

There is growing evidence that the gut microbiota affects the response to chemo- and immunotherapeutic drugs by modulating their efficacy and/or toxicity (Alexander et al., 2017; Panebianco, Andriulli, & Pazienza, 2018). Oncogenic bacteria play an important role in resistance to the chemotherapy drug gemcitabine, as they can metabolize chemotherapy drugs into inactive forms. *Clostridium difficile* is one of the most common bacterial species

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Table 1–Animals and treatment groups.

Groups	Number	Antibiotics (per os)	HEP (per os)	5-Fu (i.p.)
Control	10	/	/	/
HEP-only (FP)	10	/	100 mg/kg/day for 21 days	/
xenograft-only (Model)	10	/	/	/
pre-xenograft+HEP (PN)	10	/	Preliminary suit 5 days	Every 2 days of 5 mg/kg/day for 15 days
pre-xenograft+HEP+5-Fu (PF)	10	/		
HEP+5-Fu (NP)	10	/	100 mg/kg/day for 21 days	
5-Fu-only (NF)	10	/		
antibiotics+HEP+5-Fu (AP)	10	Preliminary suit 5 days		
antibiotics+5-Fu (AF)	10		/	

Notes: Animals were randomly divided into nine groups: control (no treatments), xenograft-only (Model), HEP-only (FP), pre-xenograft HEP (PN), pre-xenograft HEP+5-Fu (PF), HEP+5-Fu (NP), 5-Fu-only (NF), antibiotics+HEP+5-Fu (AP), and antibiotics+5-Fu (AF). Only the control and HEP-only groups had no xenografts CT-26 wt cancer cells.

Table 2–PCR primers.

Primer name	Sequence (5′→3′)	Product size (bp)
β-actin	β-actin-F1: GCTTCTAGGCGGACTGTTAC β-actin-R1: CCATGCCAATGTTGTCTCTT	100
AKT	AKT-F1: CCTGGACTACTTGCACTCCG AKT-R1: ACAGCCCGAAGTCCGTTATC	107
CCND1	CCND1-F1: GGAGCTGCTGCAAATGGAAC CCND1-R1: GCTTGTTCTCATCCGCCTCT	122
CDK4	CDK4-F1: TACATACGCAACACCCGTGG CDK4-R1: CAACTGGTCGGCTTCAGAGT	103
FOXM1	Foxm1-F1: TGGACCATTCACCCAAGTGC Foxm1-R1: GCTGTGATTCCAAGTGCTCG	103
MMP7	Mmp7-F1: GGGTTAGTTGGGGGACTGC Mmp7-R1: GACCTAGAGTGTTCCCTGGC	96
MYC	MYC-F1: GTTGGAAACCCCGCAGACA MYC-R1: GGCTGTACGGAGTCGTAGTC	90
PPAR-α	PPAR-α-F1: TCTGGGCAAGAGAATCCACG PPAR-α-R1: AAGCGTCTTCTCGGCCATAC	148
PPAR-γ	PPAR-γ-F1: CGAGGGCGATCTTGACAGG PPAR-γ-R1: ATCGAAACTGGCACCCTTGA	168

in colorectal cancer tissues in patients with reoccurrence after chemotherapy (Fukugaiti et al., 2015; Komiya et al., 2018). It can regulate the autophagy of cancer cells and make colorectal cancer resistant to chemotherapy. It can also help the cancer cells to metastasize and settle down at distant sites, which makes it a "reliable assistant" of cancer cells. Interestingly, clinical trials using the antibiotic metronidazole to kill these "reliable assistants" can effectively inhibit the proliferation of cancer cells and tumor growth. The chemotherapeutic drug cyclophosphamide was found to damage the intestinal mucous layer, allowing intestinal bacteria to enter the lymph nodes and spleen and activate specific immune cells (Daillère et al., 2016; Vétizou et al., 2015; Viaud et al., 2013). Moreover, a retrospective study of programmed cell death-1 (PD-1)/PD-ligand 1 (PD-L1) treatment in patients with advanced kidney cancer found that if the patients used broad-spectrum antibiotics a month before and after the treatment, the mean disease progression-free survival time was increased by almost fourfold (8.1 compared with 2.3 months) (Andrea et al., 2017; Patel & Crawford, 2018). In addition, there were significant differences in the efficient and total survival time. The gut microbiota may enhance the ability of immune cells to respond to tumors, but exactly which bacteria are involved in regulating immune cells remains a mystery. Animal studies indicated that additional treatment using probiotic bacteria can lower the rate of postoperative complications and adverse reactions and improve quality of life after colorectal cancer resection (Peitsidou, Karantanos, & Theodoropoulos, 2012; Serban, 2014).

In the past few decades of rapid modernization and urbanization, Chinese people's eating habits have changed dramatically (Zhang, Chang, Li, & Zhang, 2015). Dietary factors are recognized as one of the most important type of risk factor for many chronic disease outbreaks in China. Previous studies showed that diet can affect the immune response of the host, trigger inflammation, and greatly influence the composition of the gut microbiota (Verbeke, Boesmans, & Boets, 2014; Wasilewski, Zielińska, Storr, & Fichna, 2015). Moreover, different microbiota can change the metabolism of food and produce metabolites that can promote or inhibit cancer (Valdés et al., 2015). Many Chinese families have diets involving too much red meat and a lack of dietary fiber. Barbecued food-derived compounds are thought to be metabolized by the gut microbiota in metabolic by-product, associated with DNA damage (Tappel et al., 2007). Particularly, some bacteria can convert amino acids in nitrosamines and induce DNA alkanisation and mutations (Kim, Lee, Jung, & Kim, 2017; Ma, Tian, Wu, & Ma, 2017; Sasso & Latella, 2018; Zaramela et al., 2019). In contrast, a low-protein diet was shown to activate immune responses to inhibit the growth of mouse lymphoma, melanoma, and colorectal cancer (Rubio-Patiño et al., 2018), while the plant-derived protein (Bonturi et al., 2018; Møller et al., 2019), in particular, the fungal proteins from some mushrooms have been shown to have an effective immunomodulatory effect (Akanbi, Post, & van Putten, 2013; Hassan, Rouf, Tiralongo, May, & Tiralongo, 2015; Hsin et al., 2011), and improvement on gut microbiota (Chen et al., 2017). Dietary fiber can be metabolized by the gut microbiota, and the metabolites can prevent cancer; low fiber intake can cause gut microbiota disorders, leading to many diseases, including colon cancer (Bishehsari et al., 2018). Some bacteria can activate inflammation and destroy the mucous layer that protects the body from foreign invaders, creating conditions that support tumor growth (Biragyn & Ferrucci, 2018). Patients with chronic intestinal inflammation have a 2- to 10-fold increase in risk of colorectal cancer than healthy people, and nonsteroidal anti-inflammatory drugs, such as aspirin, can effectively prevent colorectal cancer (Biragyn & Ferrucci, 2018). Therefore, sensible eating is important to keep fit. To avoid the harm of too much red meat, may be the plant-derived protein will be a good alternative source of animal-derived protein. And a new report indicated that a diet rich in the amino acid histidine increases the effectiveness of methotrexate treatment and lowers toxicity in tumor mice (Kanarek et al., 2018), and rich source of amino acids in some fungus were reported benefit for healthy (Bou Zeidan et al., 2014; Kalaras, Richie, Calcagnotto, & Beelman, 2017; Pahila, Kaneda, Nagasaka, Koyama, & Ohshima, 2017).

Anticancer treatments themselves significantly affect the microbiota composition, thus disrupting homeostasis and exacerbating

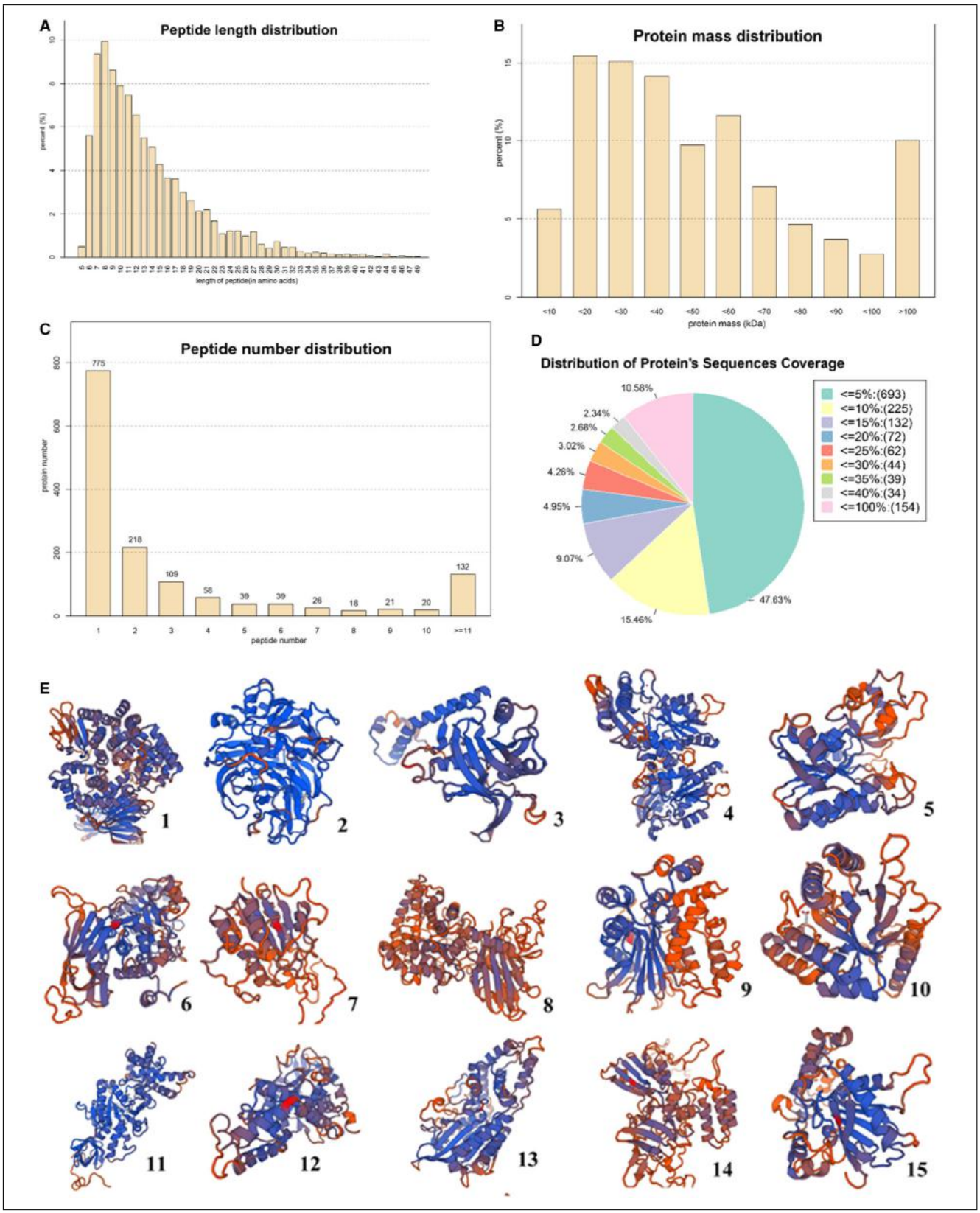

Figure 1–Protein identification of fruit bodies of *Hericium erinaceus* using liquid chromatography coupled with tandem mass spectrometry (LC-MS/MS). There are 5,568 peptides identified, of which 5,442 are unique peptide, and 1,455 proteins were identified (A and B), the MW (in kDa) most of proteins are 10 to 20 (225), 20 to 30 (220), 30 to 40 (206), 40 to 50 (142), 50 to 60 (169), 60 to 70 (103), and >100 (146) (C and D); (E) The 3D structure models of proteins from *Hericium erinaceus* were built on line in SWISS-MODEL, and there were 15 proteins showed.

A Sequencing data summary at 97% similarity

Groups	Simpson	Chao1	ACE	Shannon
FP	0.99 ± 0.0025*	1993.30 ± 198.6380*	2155.70 ± 226.3042*	8.66 ± 0.2482*
Model	0.97 ± 0.0247	1780.30 ± 396.1896♯	1885.9 2± 440.0359♯	8.06 ± 0.7419
Control	0.97 ± 0.0127	1930.56 ± 312.8154	2067.49 ± 338.9862	8.06 ± 0.4524

The richness estimators (ACE and Chao), diversity indices (Shannon and Simpson), and Coverage were calculated using the Mothur software. Values given are the mean ± SD, ♯$p < 0.05$ *vs* normal, *$P < 0.05$ *vs* model

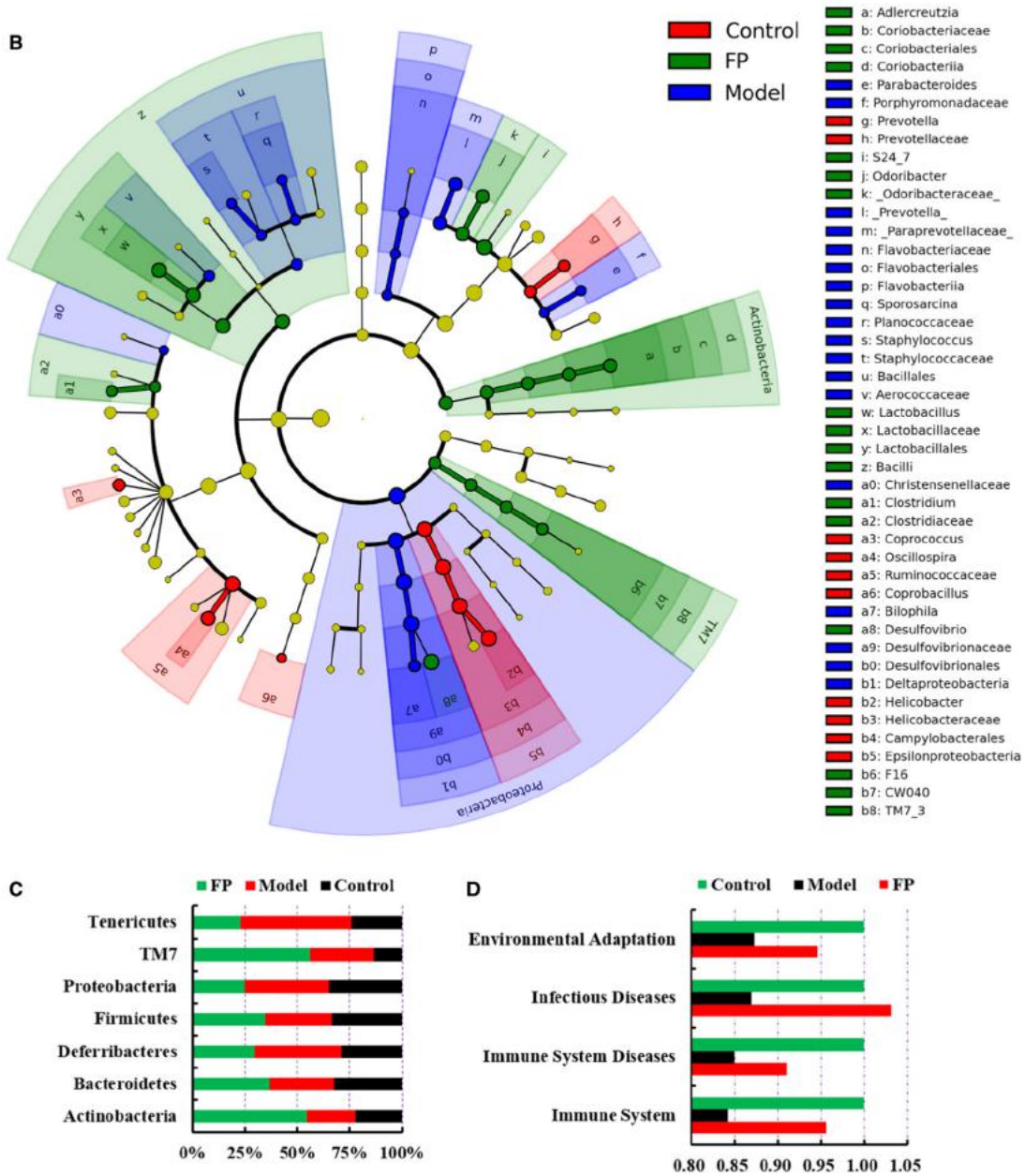

Figure 2–Effects of fungal proteins extracted from *Hericium erinaceus* on gut microbiota in normal adult mice. The graph of (A) is the sequencing data summary at 97% similarity which meaning that the alpha diversity index of proteins treated group were more abundant; (B) is the visualization of differential classification units based on classification hierarchy tree between different treated groups; (C) is the relative abundance distribution of classification units in different groups at phylum level; (D) is the KEGG pathway analysis of metagenome at level 2 using PICRUSt (Phylogenetic Investigation of Communities by Reconstruction of Unobserved States). Animals were randomly divided into three groups, Model group is the xenografted 100,000 CT 26 wt cancer cells mice; Control group is the mice with no treatment, FP is the mice with 100 mg/kg/day protein extractions lasting 21 days.

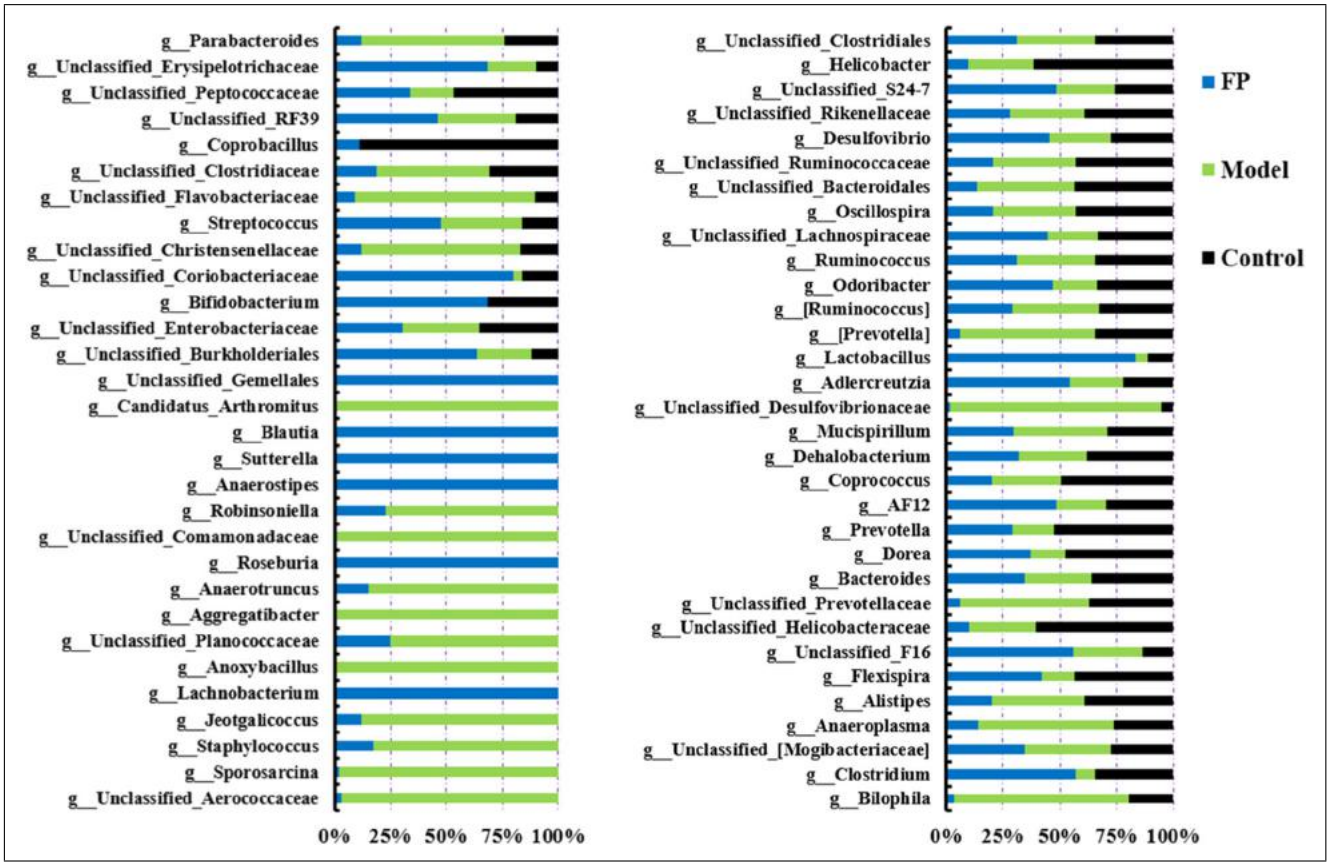

Figure 3–Effects of fungal proteins extracted from *Hericium erinaceus* on gut microbiota at genus level in normal adult mice. Animals were randomly divided into three groups, Model group is the xenografted 100,000 CT 26 wt cancer cells mice; Control group is the mice with nothing treatment, FP is the mice with 100 mg/kg/day protein extractions lasting 21 days.

patient discomfort. In this paper, the aims were to assess the auxiliary antitumor effects of fungal proteins from *Hericium erinaceus* (HEP) administered with the chemotherapy drug 5-Fluorouracil (5-Fu) to identify the role of the microbiota in mediating chemo- and immunotherapy efficacy and/or toxicity and to identify the ability of these therapeutic options to trigger dysbiosis that contributes to more severe side effects. The study also assessed an emerging dietary management strategy with fungal proteins for manipulating the microbiota in order to improve therapeutic outcomes or at least to increase quality of life during anticancer treatment.

2. MATERIALS AND METHODS

2.1 Fungal material and protein identification

Fresh whole mushroom bodies of HEP were collected from the Research Laboratory of Edible Mushrooms of Guangdong Inst. of Microbiology, China, in January 2018, and identified by Prof. Xie Yizhen of Guangdong Inst. of Microbiology.

The total proteins were extracted strictly following the instructions of a fungal protein extraction kit (BestBio BB-3127, BestBio, China), and prepared by reductive alkylation, sodium dodecyl sulfate (SDS)-polyacrylamide gel electrophoresis (PAGE), trypsin digestion, and isobaric tags for relative and absolute quantitation (iTRAQ) method. Thereafter, the protein samples were mixed and fractionated by strong cation exchange chromatography. Finally, the protein samples were quantitatively analyzed by liquid chromatography coupled with tandem mass spectrometry (LC-MS/MS). Briefly, peptides were dissolved in 0.1% Formic acid (FA), directly loaded onto a reversed-phase precolumn (Acclaim PepMap 100, Thermo, Waltham, Massachusetts). Peptide separation was performed using a reversed-phase analytical column (Acclaim PepMap RSLC, Thermo). The gradient was comprised of an increase from 6% to 22% solvent B (0.1% FA in 98% Acetonitrile (ACN)) over 26min, 22% to 35% in 8 min, and climbing to 80% in 3min then holding at 80% for the last 3 min, all at a constant flow rate of 400 nl/min on an EASY-nLC 1000 UPLC system. The peptides were subjected to NSI source followed by tandem mass spectrometry (MS/MS) in Q Exactive^{TM} plus (Thermo) coupled online to the UPLC. Intact peptides were detected in the Orbitrap at a resolution of 70,000. Peptides were selected for MS/MS using NCE setting as 28, 32; ion fragments were detected in the Orbitrap at a resolution of 17,500. A data-dependent procedure that alternated between one MS scan followed by 20 MS/MS scans was applied for the top 20 precursor ions above a threshold ion count of 10,000 in the MS survey scan with 30.0 s dynamic exclusion. The electrospray voltage applied was 2.0kV. Automatic gain control (AGC) was used to prevent overfilling of the orbitrap; 5E4 ions were accumulated for generation of MS/MS spectra. For MS scans, the m/z scan range was 350 to 1,800. Fixed first mass was set as 100 m/z. The raw data underwent peak detection, database construction, database search, quality control, and protein identification with the help of RNA-seq data of bodies.

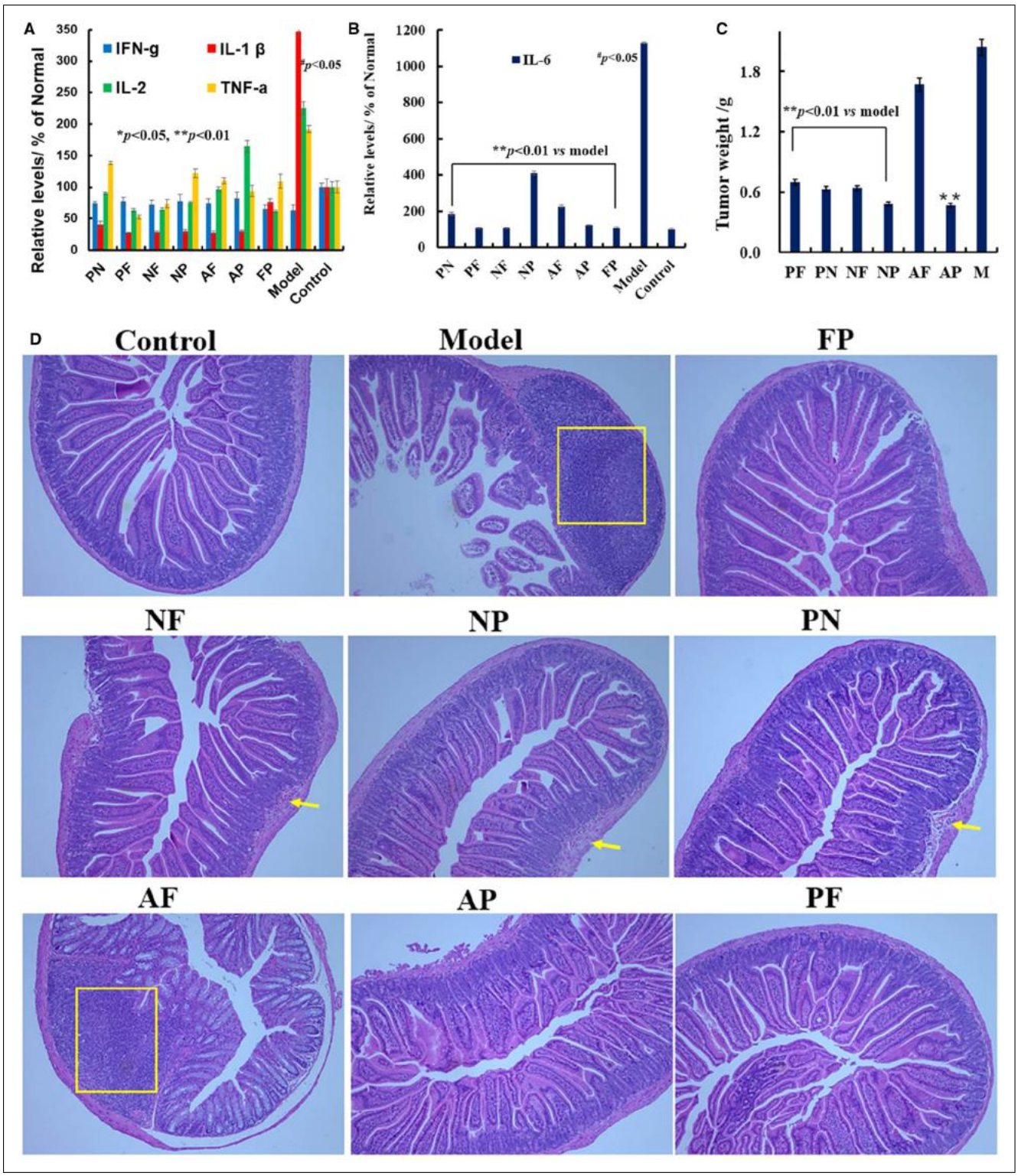

Figure 4–Effects of fungal proteins extracted from *Hericium erinaceus* on xenografted cancer mice. The graph of (A) and (B) is the level of cytokines in serum; (C) is the tumor weight of different treated groups; (D) is the colon histopathological examination using H&E staining. Animals were randomly divided into nine groups: control (no treatments), xenograft-only (Model), HEP-only (FP), pre-xenograft HEP (PN), pre-xenograft HEP+5-Fu (PF), HEP+5-Fu (NP), 5-Fu-only (NF), antibiotics+HEP+5-Fu (AP), and antibiotics+5-Fu (AF). Only the control and HEP-only groups had no xenografts. Antibiotics (ampicilin, 2 g/L; vancomycin, 1.0 g/L; neomycin, 2 g/L; metronidazole, 2 g/L). Values are expressed as means ± SDs ($n = 8$), $^{\#}P < 0.05$ compared with control group, $^{*}P < 0.05$, $^{**}P < 0.01$ compared with model group, indicates significant differences compared to the model group.

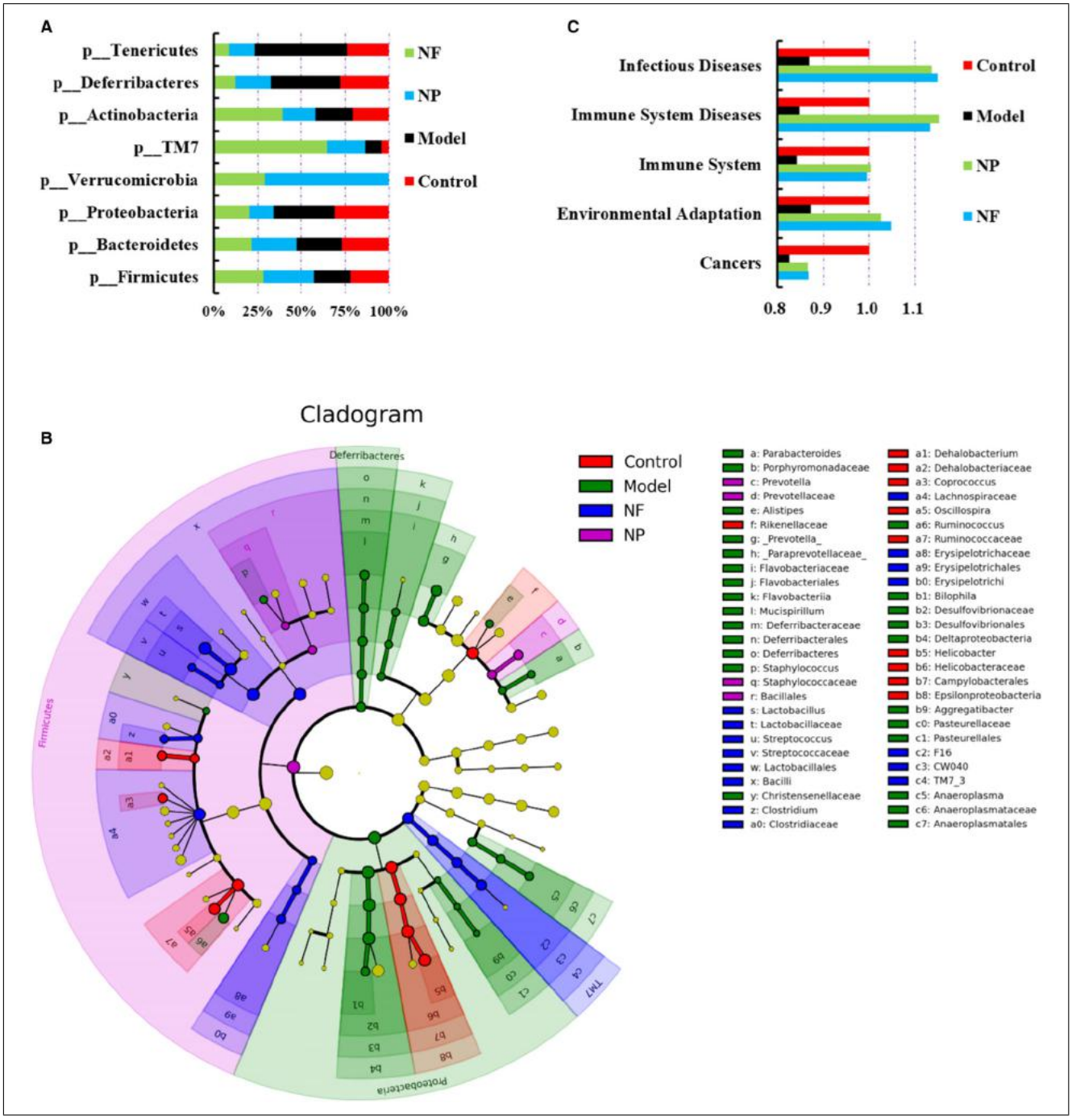

Figure 5–Effects of fungal proteins extracted from *Hericium erinaceus* on gut microbiota in xenografted cancer mice. The graph of (A) is the relative abundance distribution of classification units in different groups at phylum level; (D) is the KEGG pathway analysis of metagenome at level 2 using PICRUSt (Phylogenetic Investigation of Communities by Reconstruction of Unobserved States). Animals were randomly divided into nine groups: control (no treatments), xenograft-only (Model), HEP-only (FP), pre-xenograft HEP (PN), pre-xenograft HEP+5-Fu (PF), HEP+5-Fu (NP), 5-Fu-only (NF), antibiotics+HEP+5-Fu (AP), and antibiotics+5-Fu (AF). Only the control and HEP-only groups had no xenografts. Values are expressed as means ± SDs ($n = 8$), $^{\#}P < 0.05$ compared with control group, $^{*}P < 0.05$, $^{**}P < 0.01$ compared with model group, indicates significant differences compared to the model group.

2.2 HEP extraction

According to previous study (Chen et al., 2017), the HEP extraction was prepared. Briefly, 5 kg of mushroom bodies were ground up, and the crude proteins were extracted using alkaline extraction (pH 9 to 10) with the help of a high-pressure homogenizer and acid precipitation (pH 4.5); the precipitated matter was harvested and freeze-dried.

2.3 Animals and treatments

A total of 90 4-week-old male Balb/C mice (weighing 16 to 20 g) were purchased from the Guangdong Medical Laboratory Animal Center (GDMLAC), Guangzhou, China. The animals were kept in the specific-pathogen-free laboratory of Guangdong Laboratory Animals Monitoring Inst. at a temperature of 26 ± 1 °C and a humidity of 55 ± 10% with a 12-hr light/dark

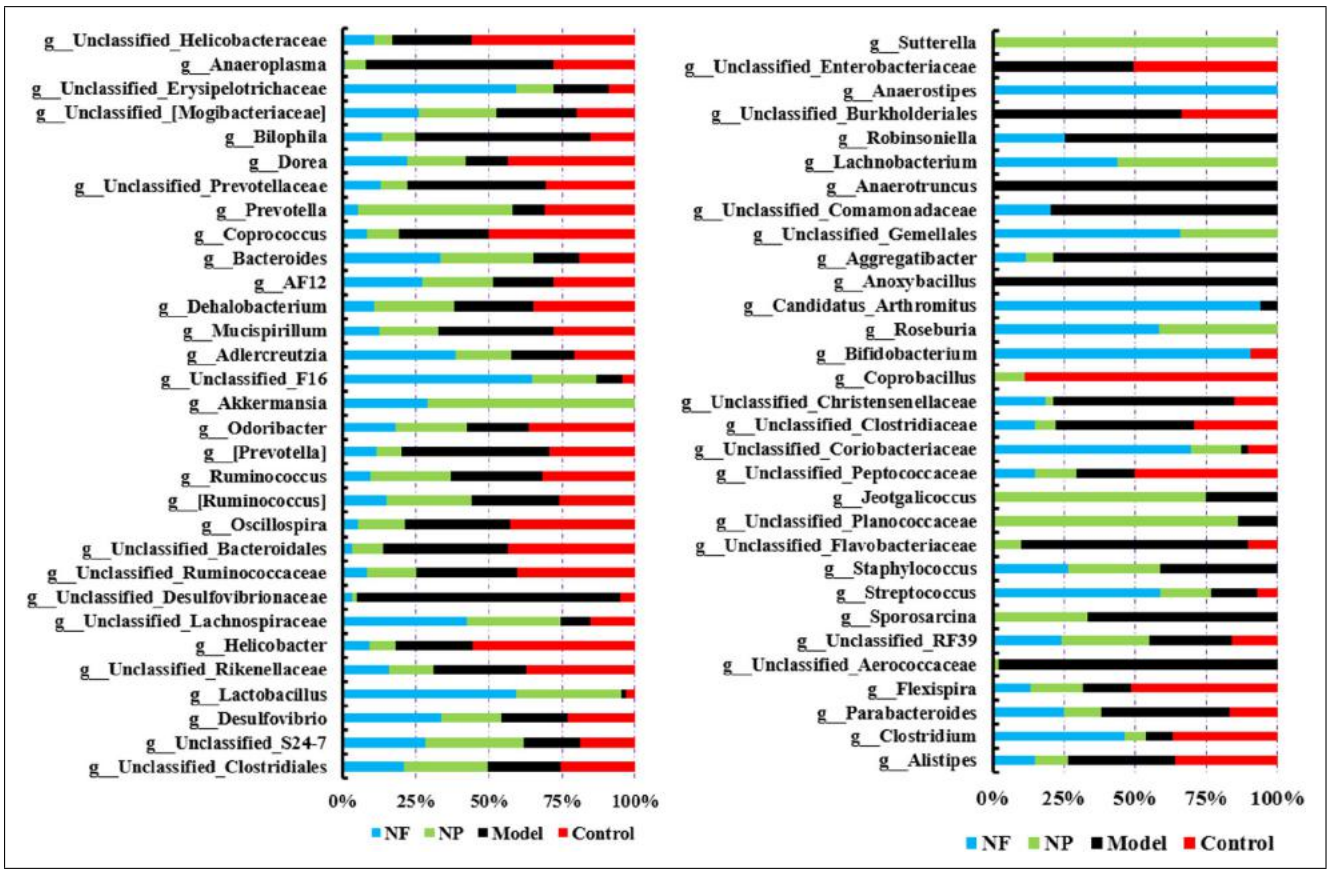

Figure 6–Effects of fungal proteins extracted from *Hericium erinaceus* on gut microbiota at genus in xenografted cancer mice. Animals were randomly divided into nine groups: control (no treatments), xenograft-only (Model), HEP-only (FP), pre-xenograft HEP (PN), pre-xenograft HEP+5-Fu (PF), HEP+5-Fu (NP), 5-Fu-only (NF), antibiotics+HEP+5-Fu (AP), and antibiotics+5-Fu (AF). Only the control and HEP-only groups had no xenografts. Values are expressed as means ± SDs ($n = 8$), $^{\#}P < 0.05$ compared with control group, $^{*}P < 0.05$, $^{**}P < 0.01$ compared with model group, indicates significant differences compared to the model group.

cycle (lights off at 17:00). The mice ate feed and drank water that had been sterilized freely. The experimental protocols were approved by the Guangdong Laboratory Animals Monitoring Inst., and all experimental procedures conformed to the Natl. Inst. of Health Guide for the Care and Use of Laboratory Animals (GT-IACUC20170420). All efforts were made to minimize the number of animals used.

After 3 days of adaptation, the 90 mice were randomly divided into nine groups, and 10 mice for each group per cage. The groups were as follows: control (no treatments), xenograft-only (Model), HEP-only (FP), pre-xenograft HEP (PN), pre-xenograft HEP+5-Fu (PF), HEP+5-Fu (NP), 5-Fu-only (NF), antibiotics+HEP+5-Fu (AP), and antibiotics+5-Fu (AF). Only the control and HEP-only groups had no xenografts. Further details of these groups are shown in Table 1.

The mice underwent xenografting by implanting 100,000 CT-26 wt cancer cells (ATCC® CRL-2638™). Briefly, cancer cells were injected hypodermically into the posterior fossa (with abundant capillaries and easy to neoplasia) after anesthesia by Diazepam (Valium™) (ip.5 mg/kg). The tumor resulting from the xenograft could be seen with the naked eye. When each tumor was the size of half a grain of rice, the xenografted mice were treated with the relevant treatments.

A total of 100 mg/kg/day of HEP was orally administrated to the mice for 21 days (for the HEP-only group), for 5 days pre-xenograft (for the pre-xenograft HEP and pre-xenograft HEP+5-Fu groups), or for 15 days post-xenograft (for the remaining four groups). For the six groups of mice treated with 5-Fu, 5 mg/kg/day 5-Fu was administered by intraperitoneal (i.p.) injection every 2 days for 14 days post-xenograft. For the two groups of mice given preliminary high-dose antibiotics (to induce dysbacteriosis), the antibiotics (ampicillin, 2 g/L; vancomycin, 1.0 g/L; neomycin, 2 g/L; metronidazole, 2 g/L, were diluted in pure water and orally administrated to mice) were administered for 5 days pre-xenograft.

2.4 Evaluation of tumor parameters

The appearance, behavior, and fur color of the mice were observed and documented daily. The mice were weighed every 3 days during the treatment period. After the treatments were finished, blood and serum samples were collected, and the tumors of the mice were dissected. Routine indices and cytokine levels were measured, including the production of interleukins (IL)-1α, IL-2, IL-6, interferon (IFN)-γ, and tumor necrosis factor (TNF)-α.

Three small intestine samples and three tumor sample from each of the nine groups were fixed in 4% paraformaldehyde and prepared as paraffin sections stained with hematoxylin-eosin (H&E) before examining using light microscopy (Chen et al., 2014; Zeng et al., 2013).

Figure 7–Effects of fungal proteins extracted from *Hericium erinaceus* on gut microbiota in xenografted cancer mice. (A-B) is the relative abundance of OTU at phylum level; (D-F) is the relative abundance of OTU at genus level in different treated groups. Animals were randomly divided into nine groups: control (no treatments), xenograft-only (Model), HEP-only (FP), pre-xenograft HEP (PN), pre-xenograft HEP+5-Fu (PF), HEP+5-Fu (NP), 5-Fu-only (NF), antibiotics+HEP+5-Fu (AP), and antibiotics+5-Fu (AF). Only the control and HEP-only groups had no xenografts. Values are expressed as means ± SDs ($n = 8$), $^{\#}P < 0.05$ compared with control group, $^{*}P < 0.05$, $^{**}P < 0.01$ compared with model group, indicates significant differences compared to the model group.

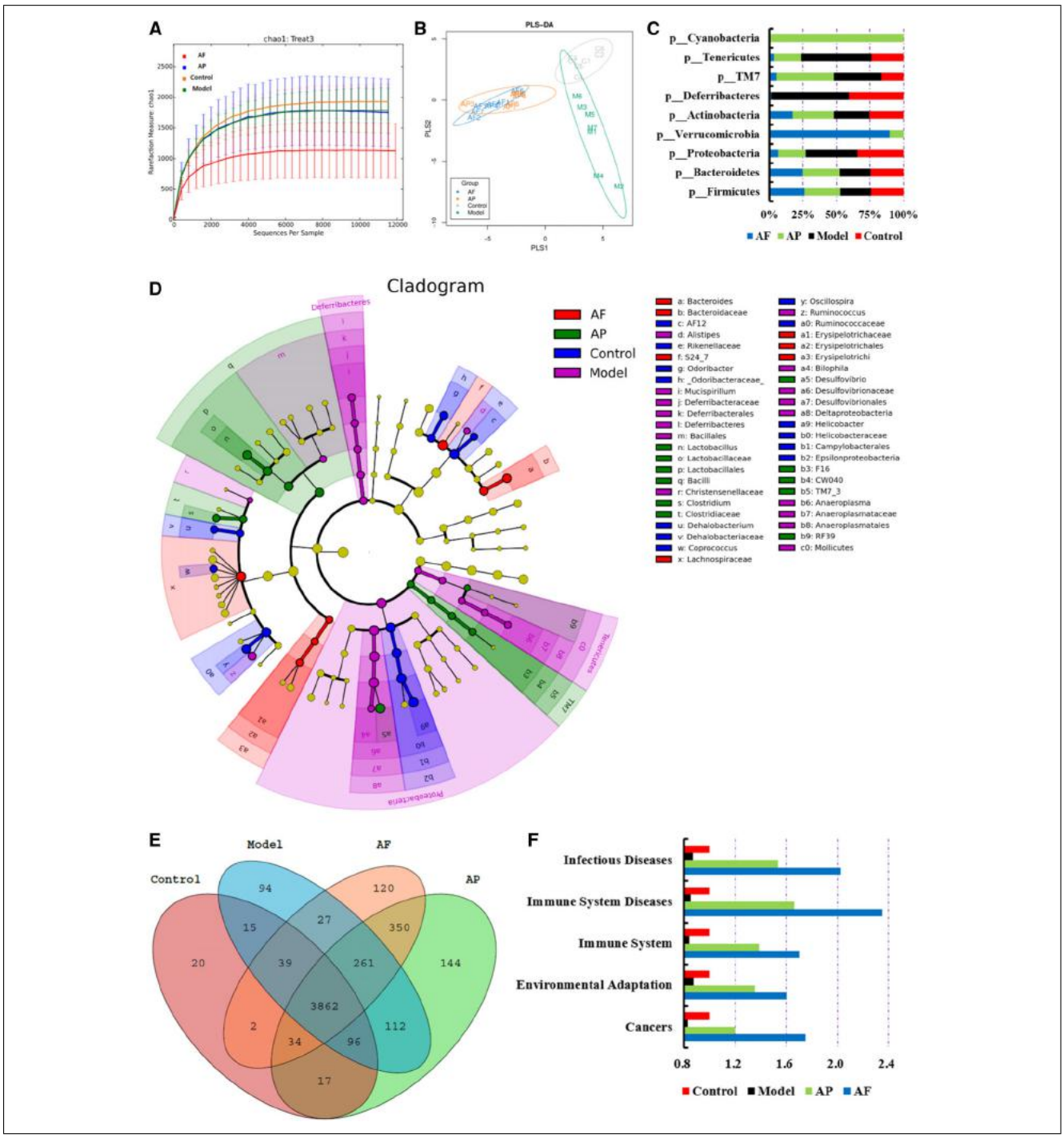

Figure 8–Effects of fungal proteins extracted from *Hericium erinaceus* on gut microbiota in dysbacteriosis xenografted cancer mice. The graph of (A) is the sequencing data summary at 97% similarity; (B) is the PCA analysis of OTU; (C) is the relative abundance of OTU at phylum level; (D) is the visualization of differential classification units based on classification hierarchy tree between different treated groups; (E) is the venn diagram of OTU in different treated groups; (F) is the KEGG pathway analysis of metagenome at level 2 using PICRUSt (Phylogenetic Investigation of Communities by Reconstruction of Unobserved States). Animals were randomly divided into nine groups: control (no treatments), xenograft-only (Model), HEP-only (FP), pre-xenograft HEP (PN), pre-xenograft HEP+5-Fu (PF), HEP+5-Fu (NP), 5-Fu-only (NF), antibiotics+HEP+5-Fu (AP), and antibiotics+5-Fu (AF). Only the control and HEP-only groups had no xenografts. Values are expressed as means ± SDs ($n = 8$), $^{\#}P < 0.05$ compared with control group, $^{*}P < 0.05$, $^{**}P < 0.01$ compared with model group, indicates significant differences compared to the model group.

2.5 Microbiome analysis

After 21 days of intervention, the mice were sacrificed and fecal samples were collected from the colonic content Frozen. Microbial DNA (with total mass of 1.2 to 20.0 ng) was isolated from each sample and preserved at –20 °C. The microbial 16S rRNA genes were amplified using the forward primer 5′-ACTCCTACGGGAGGCAGCA-3′ and the reverse primer 5′-GGACTACHVGGG TWTCTAAT-3′ for mice. Each amplified product was concentrated via solid-phase reversible immobilization and quantified by electrophoresis using an Agilent

2100 Bioanalyzer (Agilent, Santa Clara, California, USA). After quantification of the DNA concentration using NanoDrop, each sample was diluted to 1×10^9 molecules/μL in Tris-EDTA (TE) buffer and pooled. Subsequently, 20 μL of the pooled mixture was used for sequencing on an Illumina MiSeq sequencing platform according to the manufacturer's instructions. The resulting reads were analyzed as described previously (Ling et al., 2014).

2.6 Real-time (RT)-qPCR analysis

The total RNA of three small intestine samples and tumor samples from each group was extracted from the frozen tissues using Trizol reagent (Life Technologies, Carlsbad, California, USA) according to the manufacturer's instructions. The purity and concentration of the total RNA were determined using a NanoDrop spectrophotometer (ND-2100; Thermo Fisher, Waltham, Massachusetts). Using Geneseed® II First Strand cDNA Synthesis Kit (Geneseed), 1 μg of total RNA was reverse transcribed. RT-PCR was carried out in an ABI 7500HT Fast Real-Time PCR System (AB Applied Biosystems, Warrington, UK) with Geneseed qPCR SYBR® Green Master Mix and gene-specific primers. The sequences and GenBank accession numbers for the forward and reverse primers used to quantify the mRNA are listed in Table 2. The following conditions were used for RT-PCR: 95 °C for 5 min, then 95 °C for 10 s, and 60 °C for 34 s in 40 cycles, followed by 95 °C for 15 s, 60 °C for 1 min, and 95 °C for 15 s. Each sample was assessed in triplicate for all primers.

2.7 Western blotting analysis

Proteins were extracted from the tumor and colon tissue samples in radioimmunoprecipitation assay (RIPA) lysis buffer containing phenylmethylsulfonyl fluoride (PMSF; P0013B; Beyotime, Shanghai, China). Proteins were separated by SDS-PAGE (8% to 15%) and electrophoretically transferred onto polyvinylidene fluoride membranes. The membranes were blocked with 5% nonfat dry milk in Tris-buffered saline with Tween-20 (Aladdin, T104863, Shanghai, China) for 60 min, probed with primary antibodies overnight at 4 °C and then incubated with goat antimouse IgG secondary antibody (1:2000). The primary antibodies targeted the following molecules: MMP7 (AF0218, Proteintech Group, Chicago, Illinois, USA), TNF-α (AF7014, Proteintech Group, Chicago, Illinois, USA), MYC (AF0358, Proteintech Group, Chicago, Illinois, USA), PTEN (AF6351, Proteintech Group, Chicago, Illinois, USA), Akt (ab151279, Abcam, UK), CCDN1 (AF0931, Proteintech Group, Chicago, Illinois, USA), CKD4 (DF6102, Affinity Biosciences, OH, USA), FOXM1 (DF6962, Affinity Biosciences, OH, USA), PPAR-α (AF6284, Proteintech Group, Chicago, Illinois, USA), and PPAR-γ. Band intensity was quantified using ImageJ software (NIH, Bethesda, Maryland, USA).

2.8 Histopathological examinations

The colon cancer tissues were removed and fixed in the 4% paraformaldehyde at pH 7.4 for further pathological observation. These tissue samples were made into paraffin sections after drawing materials, fixation, washing, dehydration, transparency, dipping in wax, and embedding. The colon and tumor of animals were dissected. A total of four samples from each group were fixed in 4% paraformaldehyde solution and prepared as paraffin sections. Sections were stained with H&E. Immunostaining using paraffin-embedded 3 μm sections and a two-step peroxidase conjugated polymer technique (DAKO Envision kit, DAKO, Carpinteria, CA, USA). Slides were observed by light microscopy (Chen et al., 2017). They were also prepared for immunohistochemistry analysis, using primary antibodies against TNF-α (1:300), PPAR-α (1:150), and FOXM1 (1:200).

2.9 Statistical analysis

All data are described as the means ± standard deviations (SD) of at least three independent experiments. The significant differences between treatment groups were analyzed by one-way analysis of variance (ANOVA) at $P < 0.05$ using SPSS (Abacus Concepts, Berkeley, CA, USA) and Prism5 software (GraphPad, San Diego, CA, USA).

3. RESULTS AND DISCUSSION

3.1 *H. erinaceus* mushroom bodies protein identification

There were 117838 LC-MS/MS spectra in total, among which 41,291 were unique. A total of 5,568 peptides were identified, of which 5,442 were unique, and 1,455 proteins were identified. The molecular weight (in kDa) of most of the proteins were 10 to 20 (225), 20 to 30 (220), 30 to 40 (206), 40 to 50 (142), 50 to 60 (169), 60 to 70 (103), and >100 (146). More details of the peptides and proteins are shown in Figure 1, and 3D models of some were built using SWISS- MODEL.

3.2 Effects of HEP on the gut microbiota in adult mice

The effects of HEP on the gut microbiota in adult mice are shown in Figure 2. The alpha diversity index based on the sequencing data (using a threshold of 97% similarity) showed that the HEP-only group could improve the microbial richness and diversity compared to the xenograft-only group and even compared to the control group (Figure 2A). The variance analysis involving linear discriminant analysis effect size (LEfSe) in Figure 2B showed that HEP and the transplanted tumor influenced the gut microbiota. The relative abundance distribution of classification units at the phylum level in different treatment groups is shown in Figure 2C. Depth analysis indicated that HEP favored the microbial functional pathways accounting for immune system, immune system diseases, infectious diseases, and environmental adaptation based on the Kyoto Encyclopedia of Genes and Genomes (KEGG) pathway metagenome analysis at level 2 using Phylogenetic Investigation of Communities by Reconstruction of Unobserved States (PICRUSt) (Figure 2D). The changes in the gut microbiota at genus level indicated that the pathogenic bacteria were reduced and some probiotic bacteria were increased, as shown in Figure 3. Certain aerobic and microaerobic bacteria were increased in the xenograft-only group, such as *Parabacteroides*, *Flavobacteriaceae*, *Christensenellaceae*, *Robinsoniella, Comamonadaceae*, *Anaerotruncus*, *Candidatus_Arthromitus*, *Aggregatibacter*, *Planococcaceae*, *Anoxybacillus*, *Jeotgalicoccus*, *Desulfovibrionaceae*, *Sporosarcina*, *Staphylococcus*, *Aerococcaceae*, and *Bilophila* (Figure 3). In contrast, HEP led to increases in some probiotic bacteria such as the relative abundance of *Bifidobacterium*, *Gemellales*, *Blautia*, *Sutterella*, *Anaerostipes*, *Roseburia*, *Lachnobacterium*, *Lactobacillus*, and *Desulfovibrio* (Figure 3). The statistical analysis was performed on the basis that one abundance of Operational Taxonomic Unit (OTU) was defined as one functional unit. The liquid protein chip analysis of the IFN-γ, IL-1β, IL-2, TNF-α, and IL-6 levels in serum samples (Figure 4A and 4B) showed that the anti-inflammatory factors were activated and the pro-inflammatory factors were inhibited. These results are consistent with our previous results (Chen et al., 2017) that showed the HEP triggered functional pathways accounting for immune activity, resistance to infection, and environmental adaptation (Figure 2D).

A Sequencing data summary at 97% similarity

Group	Simpson	Chao1	ACE	Shannon
PF	0.98 ± 0.0090	2013.65 ± 392.2712	2180.24 ± 431.9247	8.24 ± 0.4351
PN	0.98 ± 0.0218	1851.97 ± 300.5410	1981.13 ± 361.1806	8.38 ± 0.4901
Model	0.97 ± 0.0247	1780.30 ± 396.1896	1885.92 ± 440.0359	8.06 ± 0.7419
Control	0.97 ± 0.0127	1930.56 ± 312.8154	2067.49 ± 338.9862	8.06 ± 0.4524

The richness estimators (ACE and Chao), diversity indices (Shannon and Simpson), and Coverage were calculated using the Mothur software. Values given are the mean ± SD, #$p < 0.05$ *vs* normal, *$p < 0.05$ *vs* model

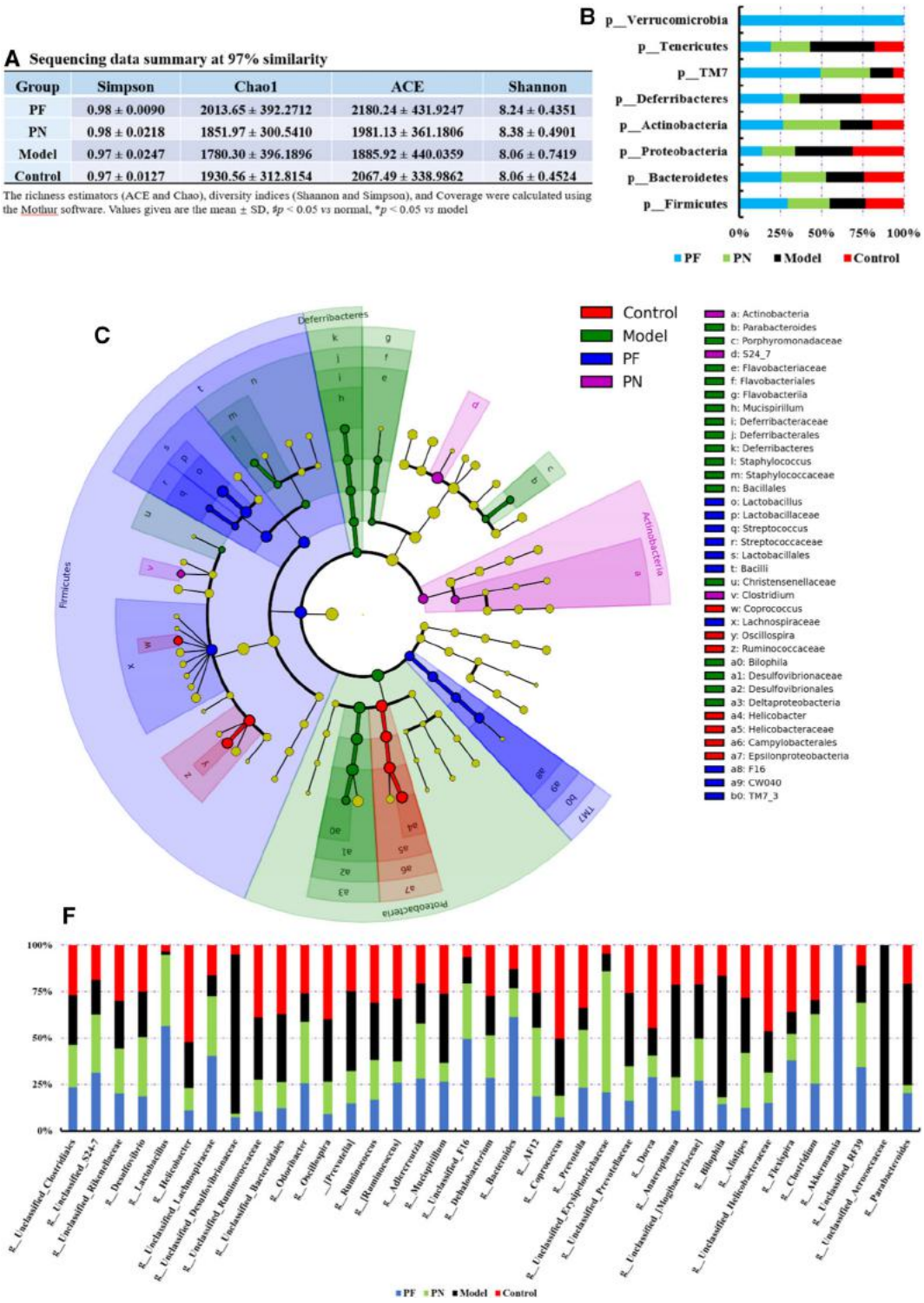

Figure 9–Effects of fungal proteins extracted from *Hericium erinaceus* on gut microbiota in xenografted cancer mice. The graph of (A) is the sequencing data summary at 97% similarity; (B) is the relative abundance of OTU at phylum level; (C) is the visualization of differential classification units based on classification hierarchy tree between different treated groups; (F) is the relative abundance of OTU at genus level in different treated groups. Animals were randomly divided into nine groups: control (no treatments), xenograft-only (Model), HEP-only (FP), pre-xenograft HEP (PN), pre-xenograft HEP+5-Fu (PF), HEP+5-Fu (NP), 5-Fu-only (NF), antibiotics+HEP+5-Fu (AP), and antibiotics+5-Fu (AF). Only the control and HEP-only groups had no xenografts. Values are expressed as means ± SDs ($n = 8$), #$P < 0.05$ compared with control group, *$P < 0.05$, **$P < 0.01$ compared with model group, indicates significant differences compared to the model group.

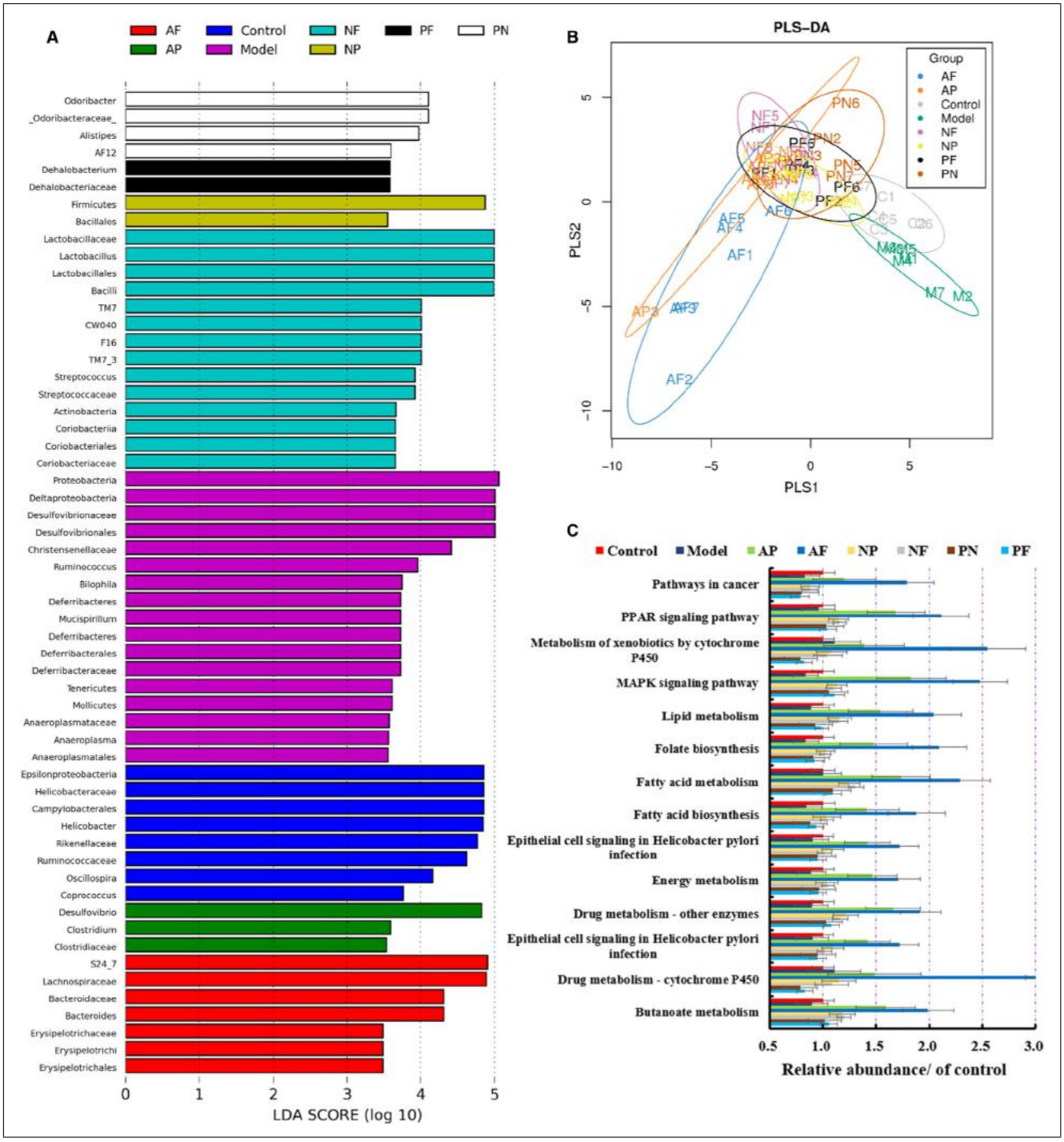

Figure 10–Effects of fungal proteins extracted from *Hericium erinaceus* on gut microbiota in normal and xenografted cancer mice. (A) is the LEfSe analysis of differential classification units between different treated groups; (B) is the PCA analysis of OTU; (C) is the KEGG pathway analysis of metagenome at level 2 using PICRUSt (Phylogenetic Investigation of Communities by Reconstruction of Unobserved States). Animals were randomly divided into nine groups: control (no treatments), xenograft-only (Model), HEP-only (FP), pre-xenograft HEP (PN), pre-xenograft HEP+5-Fu (PF), HEP+5-Fu (NP), 5-Fu-only (NF), antibiotics+HEP+5-Fu (AP), and antibiotics+5-Fu (AF). Only the control and HEP-only groups had no xenografts. Values are expressed as means ± SDs ($n = 8$), $^{\#}P < 0.05$ compared with control group, $^{*}P < 0.05$, $^{**}P < 0.01$ compared with model group, indicates significant differences compared to the model group.

3.3 Auxiliary effects of HEP on the gut microbiota in mice with xenografted tumors

To evaluate the auxiliary antitumor effects of HEP, we implanted CT-26 wt cancer cells into the posterior fossa of the mice. When each tumor was the size of half a grain of rice, one group of xenografted mice were treated with 100 mg/kg/day HEP & i.p. 5 mg/kg/day 5-Fu every 2 days for 15 days (HEP+5-Fu group), and another group was treated with i.p. 5 mg/kg/day of 5-Fu every 2 days for 15 days (5-Fu-only group). As shown in Figure 4C, the tumor weight was lower in the 5-Fu-only group than in the

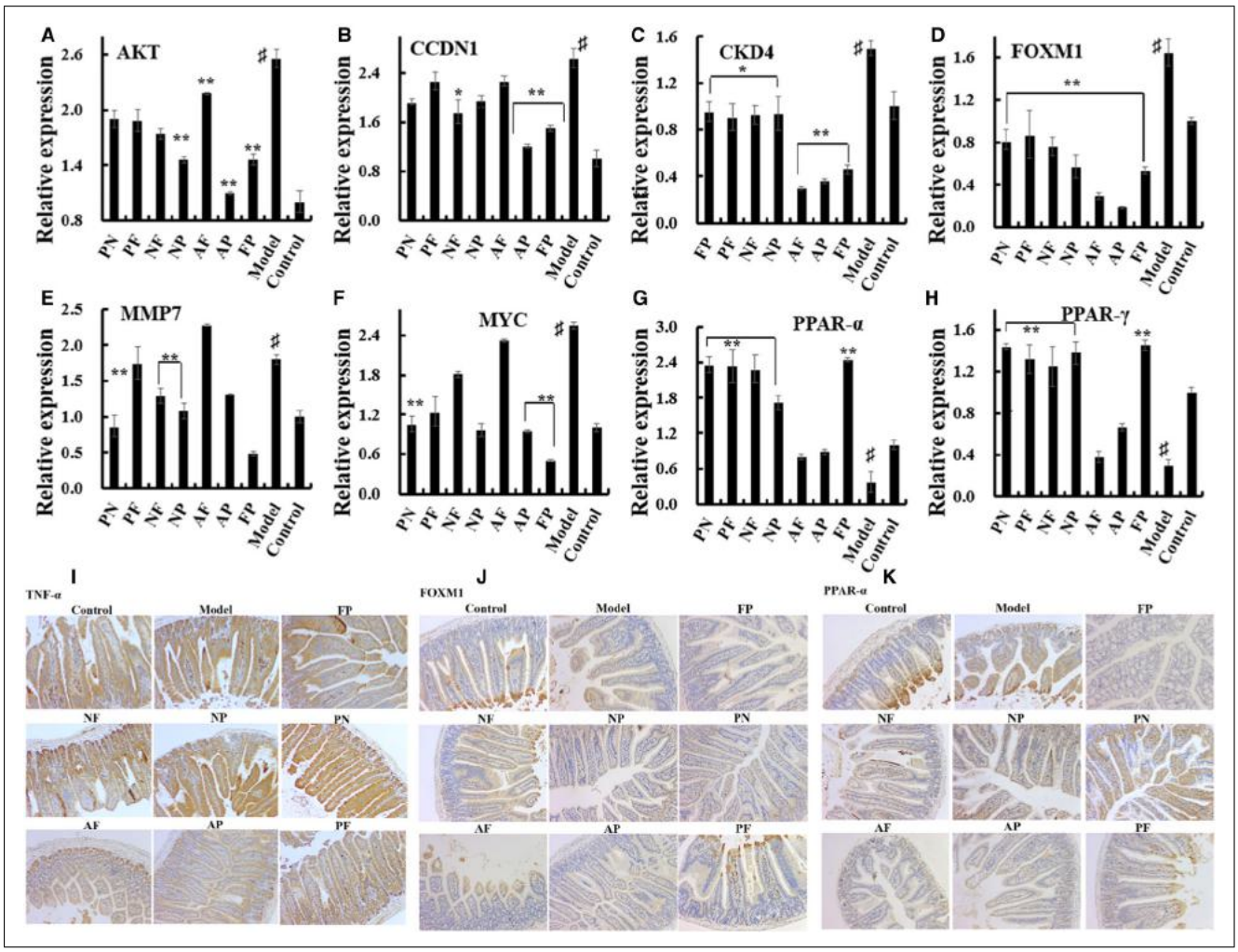

Figure 11–Effects of fungal proteins extracted from *Hericium erinaceus* on colon tissues in xenografted cancer mice. (A-H) is the QT-PCR analysis results of some key genes in the colon tissues; (I) (TNF-α); (J) (FOXM1); and (K) (PPAR-α) is the immunohistochemical staining of the colon tissues. Animals were randomly divided into nine groups: control (no treatments), xenograft-only (Model), HEP-only (FP), pre-xenograft HEP (PN), pre-xenograft HEP+5-Fu (PF), HEP+5-Fu (NP), 5-Fu-only (NF), antibiotics+HEP+5-Fu (AP), and antibiotics+5-Fu (AF). Only the control and HEP-only groups had no xenografts. Values are expressed as means ± SDs ($n = 8$), $^{\#}P < 0.05$ compared with control group, $^{*}P < 0.05$, $^{**}P < 0.01$ compared with model group, indicates significant differences compared to the model group.

xenograft-only group ($P < 0.05$), and the addition of HEP led to a significantly lower tumor size than that in the 5-Fu-only group ($P < 0.05$), which showed that HEP can accelerate the antitumor effects of 5-Fu.

Bacteria have been shown to modulate chemotherapy responses (Biragyn & Ferrucci, 2018; Kanarek et al., 2018). Also, anticancer treatments themselves significantly affect the microbiota composition (Biragyn & Ferrucci, 2018; Kanarek et al., 2018), thus disrupting homeostasis and exacerbating patient discomfort. 5-Fu, a chemotherapy drug, changed the microbiota composition (Figure 5C), as the abundances of the phyla Actinobacteria and TM7 were highly imbalanced (Figure 5A). More details (at genus level) are shown in Figure 6, which shows that the functional units of *Erysipelotrichaceae*, *F16*, *Anaerostipes*, *Gemellales*, *Arthromitus*, *Coriobacteriaceae*, and *Streptococcus* in the 5-Fu-only group were increased more than in the control group. Certain probiotic bacteria, such as *Roseburia*, *Lactobacillus*, *Bifidobacterium*, and *Akkermansia* were increased in the 5-Fu-only group, which suggests that 5-Fu could improve the abundance of certain probiotic bacteria to activate the immune system and thereby inhibit tumor growth (Figure 4). However, 5-Fu promoted some bacteria overgrowth (Figure 6 and 7). Fortunately, we found that HEP has synergistic effects that can improve the microbial diversity and homeostasis to reduce the toxic side effects of 5-Fu (Figure 3, 5, 6 and 8), though the mechanisms need more research.

3.4 Auxiliary effects of HEP on the gut microbiota in mice with xenografted tumors and dysbacteriosis

To increase our understanding of the role of the gut microbiota in tumor responses to 5-Fu, before carrying out the xenograft, the mice were preliminary treated with antibiotics (to induce dysbacteriosis) for 5 days (ampicillin, 2 g/L; vancomycin, 1.0 g/L; neomycin, 2 g/L; and metronidazole, 2 g/L). When each tumor was the size of half a grain of rice, a group of mice was treated with 100 mg/kg/day HEP & i.p. 5 mg/kg/day of 5-Fu every 2 days for 15 days (antibiotics+HEP+5-Fu group), and another group was treated with i.p. 5 mg/kg/day of 5-Fu only every 2 days for 15 days (antibiotics+5-Fu group). The tumor weight in the antibiotics+5-Fu group was higher than that in the antibiotics+HEP+5-Fu group ($P < 0.05$), and close to that in the xenograft-only group (Figure 4C), which indicated that the antitumor effect of 5-Fu almost failed. A total of 16S rRNA sequence

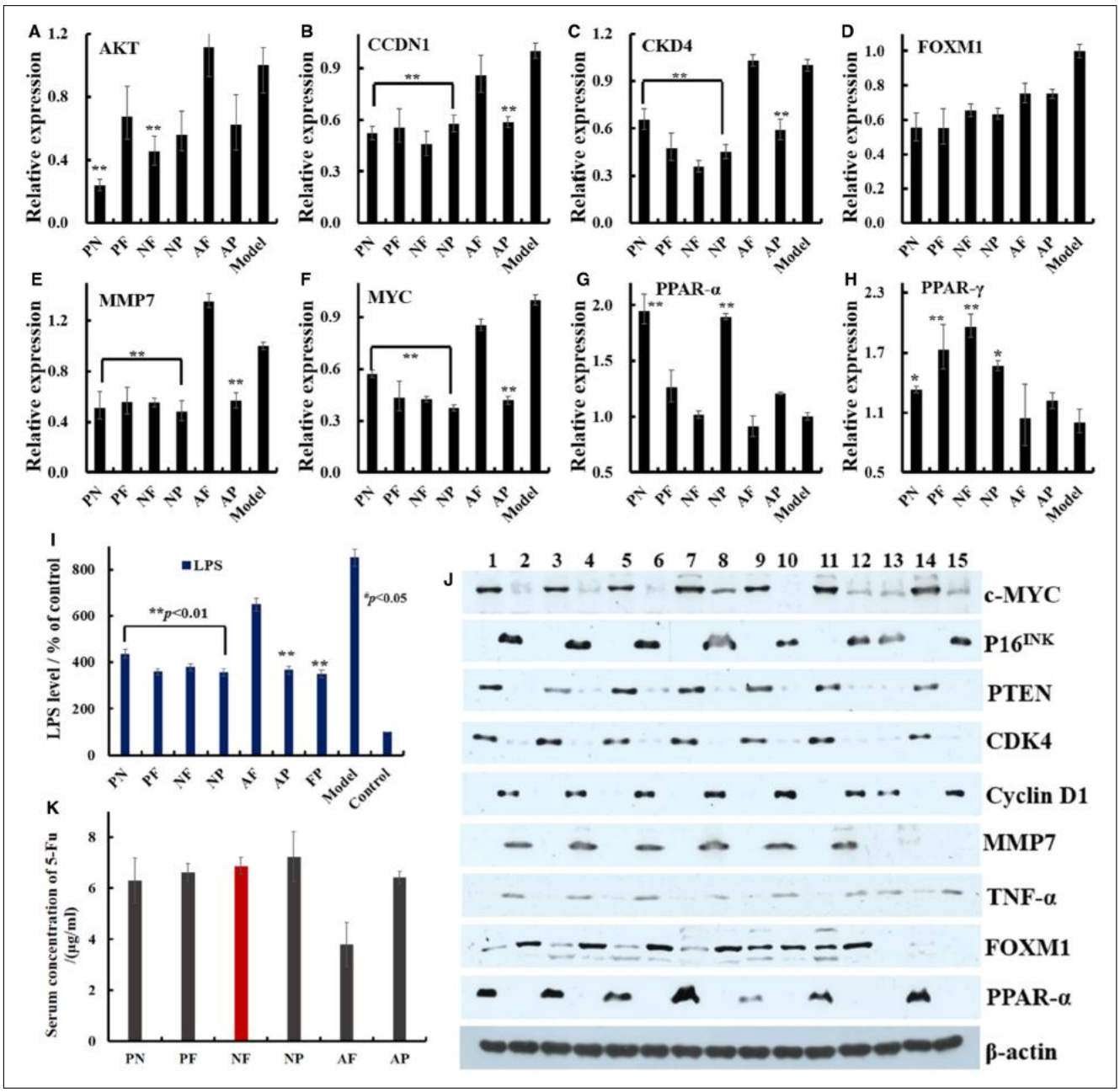

Figure 12–Effects of fungal proteins extracted from *Hericium erinaceus* on tumor tissues in xenografted cancer mice. (A-H) is the QT-PCR analysis results of some key genes in the tumor tissues; (I) is the western bolting analysis results of some key proteins in the tumor tissues; (J) is the level of LPS in serum; (K) is the plasma concentration of 5-Fu. Animals were randomly divided into nine groups: control (no treatments), xenograft-only (Model), HEP-only (FP), pre-xenograft HEP (PN), pre-xenograft HEP+5-Fu (PF), HEP+5-Fu (NP), 5-Fu-only (NF), antibiotics+HEP+ 5-Fu (AP), and antibiotics+5-Fu (AF). Only the control and HEP-only groups had no xenografts. Values are expressed as means ± SDs ($n = 8$), $^{\#}P < 0.05$ compared with control group, $^{*}P < 0.05$, $^{**}P < 0.01$ compared with model group, indicates significant differences compared to the model group.

analysis provided more evidence, showing that the abundances of most bacteria in the antibiotics+5-Fu groups were changed more than in the 5-Fu-only group (Figure 7C), as the TM7, Deferribacteres, Actinobacteria, and Proteobacteria decreased, while Verrucomicrobia increased; and the changes in microbiota composition at genus level, as the abundance of *S24-7*, *Lachnospiraceae*, *Akkermansia, Lactobacillus*, *Dorea*, *Blautia*, *Bacteroides*, *Roseburia*, *Enterobacter*, *Enterococcus*, *Anaerotruncus, Proteus*, and *Bifidobacterium* were exceedingly increased (Figure 7F). All the results demonstrated that the antitumor effect of 5-Fu might rely on the gut microbiota, that is, improvements in the gut microbiota composition can significantly promote the antitumor effect of 5-Fu.

The administration of HEP significantly improved both the 5-Fu and the antibiotic-induced dysbacteriosis in mice, showing better effects in those mice treated with the antibiotics. All the results indicated that the antitumor effect of 5-Fu relies on gut microbiota participation. Therefore, adding prebiotics and/or synbiotics may promote tumor responses to 5-Fu, and/or adding probiotic bacteria before chemotherapy may be a good oncotherapy, though clinical trials are needed to confirm these hypotheses.

3.5 Prophylactic effects of HEP against tumors

After showing that HEP has auxiliary antitumor effects, we also wanted to know whether it has prophylactic antitumor effects. Thus, mice were preliminary treated with 100 mg/kg/day HEP for 5 days before carrying out the xenograft. As shown in Figure 4C, the tumor weight in the pre-xenograft HEP group was lower than that in the pre-xenograft HEP+5-Fu group. However, a comparison of the HEP+5-Fu and 5-Fu-only groups showed that the difference in tumor weight was not significant ($P > 0.05$). According to the predicted functional pathways (Chen et al., 2017), the HEP induced the microbial functions accounting for immune activity, resistance to infections, and environmental adaptation (Figure 2D, 5C, 7F, and 10C). The above-mentioned effects were demonstrated via modulations of the gut microbiota composition (Figure 9B), as the most bacteria were improved (Figure 9B and 9F), such as the high abundance of the four species (*Desulfovibvionaceae*, *Aerococcaceae*, *Bilophila*, and *Parabacteroides*) in the model group, which changed significantly. All the results demonstrated that HEP has auxiliary and prophylactic antitumor effects.

3.6 Auxiliary antitumor mechanism of HEP

Research has shown that many nonantibiotic drugs can inhibit intestinal bacteria, including some nonantibiotic drugs with broad-spectrum effects (Spanogiannopoulos & Turnbaugh, 2018), which means that drug side effects can directly affect intestinal bacteria cells and may lead to eventual drug resistance or even an immediate effect on efficacy. The toxicity of nonantibiotic drugs to the host and also the intestinal bacteria needs to be considered in future drug development. In this paper, we found that 5-Fu exacerbates dysbacteriosis, which causes its antitumor effects to be reduced (Figure 3 and 7). The auxiliary and prophylactic effects of HEP on the gut microbiota in mice with xenografted tumors are shown in Figure 10, in which the bacterial compositions of different treatment groups were distinguished using LEfSe (Figure 10A) and principal component analysis (PCA) (Figure 10B) in all groups. The abundance of some probiotic bacteria were increased, including *Roseburia*, *Lactobacillus*, *Bifidobacterium*, and *Akkermansia*, while some aerobic and microaerobic bacteria were inhibited, such as *Parabacteroides*, *Flavobacteriaceae*, *Christensenellaceae*, *Robinsoniella, Comamonadaceae*, *Candidatus_Arthromitus*, *Anaerotruncus*, *Aggregatibacter*, *Anoxybacillus*, *Planococcaceae*, *Desulfovibrionaceae*, *Jeotgalicoccus*, *Sporosarcina*, *Staphylococcus*, *Aerococcaceae*, and *Bilophila*. The levels of some pro-inflammatory factors were also inhibited (Figure 3A and 3B). Polymerase Chain Reaction (PCR) results (Figure 11A) showed that the expressions of Akt, CCDN1, CKD4, FOXM1, MMP7, MYC, PPAR-α, and PPAR-γ were improved in the HEP+5-Fu groups, and immunohistochemistry results (Figure 11B) showed that the level of PPAR-γ was upregulated and TNF-α was downregulated in the HEP+5-Fu groups. In addition, the levels of lipopolysaccharide (LPS; Figure 12I) were reduced ($P < 0.05$). These results all indicated that internal homeostasis was improved after HEP treatment, which means that the auxiliary and prophylactic antitumor effects of HEP may function by regulating the gut microbiota compositions and thereby controlling the immune response.

To increase our understanding of the auxiliary antitumor molecular mechanism of HEP, we also reviewed colon cancer transcriptome sequencing data of a paper and a database (Schirripa & Lenz, 2016), combined with predictive analytics results of metagenomics at level 3 (Figure 10C), which showed that HEP mainly regulates the PPAR signaling pathway, mitogen-activated protein kinase (MAPK) signaling pathway, pathway in cancer, and lipid, fatty acid, drug, and energy metabolism. We used the RT-PCR and western blotting results to identify key genes and proteins in the colon cancer tissues. The results in Figure 12 show that the expression of MMP7, MYC, PTEN, Akt, CCDN1, CKD4, FOXM1, PPAR-α, and PPAR-γ were significantly improved ($P < 0.05$).

3.7 Discussion

The gut microbiota is affected by drug dosages, pharmaceutical compositions, length of treatment course, and adjuvant use (Noh et al., 2017). At the same time, the changes in the gut microbiota may change the pharmacokinetic characteristics of the therapeutic drug. Microbial enzymes can activate or deactivate drug effects and alter patient responses to drugs by affecting liver function. Dietary and microflora interventions are expected to enhance drug efficacy, reduce adverse effects of drugs, and even change the metabolic characteristics of drugs. Previous studies have showed that the probiotics administration improve the imbalance of gut microbiota in mice, and which contributed to the reduction of antitumor efficacy of 5-FU (Yuan et al., 2018). And gut microbiota can modulate or restore the intestinal immune response in the context of 5-fu induced mucositis (Li et al., 2017). Studies also provided a number of solutions to improve the gut microbiota to modulate the chemotherapy efficacy and toxicity (Alexander et al., 2017), as Carboxymethyl pachyman (CMP) can reduce intestinal mucositis and regulate the intestinal microflora in 5-fluorouracil-treated CT26 tumour-bearing mice (Wang, Yang, Gao, Wang, & Cao, 2018); volatile oil from *Amomi fructus* attenuates 5-Fluorouracil induced intestinal mucositis (Zhang et al., 2017); *lactobacillus casei* variety rhamnosus probiotic preventively attenuates 5-Fluorouracil/Oxaliplatin-induced intestinal injury in a syngeneic colorectal cancer model (Chang et al., 2018); and combination therapy of *Lactobacillus plantarum* supernatant and 5-Fluouracil increases chemosensitivity in colorectal cancer cells (An & Ha, 2016). In this paper, animal research showed that HEP with 5-Fu can significantly suppress tumor growth and inhibit inflammatory markers such as IFN-γ, IL-1β, IL-2, TNF-α, IL-6, and LPS. 16S rRNA sequencing showed that HEP can ameliorate the dysbacteriosis induced by 5-Fu, as some aerobic and microaerobic bacteria in the mice with xenografted tumors were inhibited, including *Parabacteroides*, *Flavobacteriaceae*, *Christensenellaceae*, *Robinsoniella, Comamonadaceae*, *Candidatus_Arthromitus*, *Anaerotruncus*, *Aggregatibacter*, *Anoxybacillus*, *Planococcaceae*, *Desulfovibrionaceae*, *Jeotgalicoccus*, *Sporosarcina*, *Staphylococcus*, *Aerococcaceae*, and *Bilophila*, while some probiotic bacteria were increased, such as *Bifidobacterium*, *Gemellales*, *Anaerostipes*, *Roseburia*, *Lachnobacterium*, *Lactobacillus*, *Blautia, Desulfovibrio*, and *Sutterella*. The PCR and Western blotting results also showed that the expressions of oncogenes MMP7 and MYC were inhibited after treatment with HEP with 5-Fu, and the expressions of Akt, CCDN1, CKD4, FOXM1, PPAR-α, and PPAR-γ were increased.

LPS, so-called endotoxin, is a component of the cell walls of Gram-negative bacteria. It has thermal stability and low antigenicity and is toxic to the host. The main toxic ingredient is lipid A. It is only released after bacterial death and disintegration, or it can be released from bacterial cells by artificial methods. It can cause fever, microcirculatory disturbances, endotoxic shock, disseminated intravascular coagulation, and even the progression of colorectal cancer (Blaut, 2015; Zhu et al., 2016). In this study, certain Gram-negative bacteria increased in the mice with xenografted tumors. For example, *Desulfovibrionaceae*, *Bilophila*, *Porphyromonadaceae*, and *Flavobacteriaceae* were significantly increased so that the abundance

levels in the xenograft-only group were 17.7-, 3.94-, 2.64- and 7.61-fold, respectively, by comparing to the levels in the control group ($P < 0.01$). After treatment with 5-Fu, the levels of LPS secretion were completely inhibited, and the addition of HEP led to further bacterial inhibition; accordingly, the levels of LPS were lower and the suppression of tumor growth was much improved (Figure 4 and 12I). This demonstrated that HEP is a good antimicrobial for improving the gut microbiota. These results are consistent with our previous findings that HEP3 can inhibit the proliferation of certain Gram-negative bacteria such as *Bacteroides* spp., *B. vulgatus*, and *Desulfovibrio* spp. (Chen et al., 2017).

The family *Erysipelotrichaceae* belongs to the phylum Firmicutes, and its presence in healthy and/or diseased hosts is on the rise (Kaakoush, 2015). The importance of *Erysipelotrichaceae* in inflammation-related disorders of the gastrointestinal tract is highlighted by the fact that it has been found to be enriched in colorectal cancer, as its abundance levels were found to be increased in the lumen of colorectal cancer patients (Chen, Liu, Ling, Tong, & Xiang, 2012) and in animals with 1,2-dimethylhydrazine-induced colon cancer (Zhu et al., 2014). Members of this bacterial family appear to be highly immunogenic and can potentially flourish after treatment with broad-spectrum antibiotics. The relative abundance of *Erysipelotrichi* positively correlated with TNF-α levels (Dinh et al., 2015), and the levels of *Erysipelotrichaceae* incertae sedis in mice were increased after treatment with high doses of gentamicin (Zhao et al., 2013). In this study, the abundance level of *Erysipelotrichaceae* (Unclassified_*Erysipelotrichaceae* and *Coprobacillus*) in the xenograft-only group was higher than in the control group, and *[Eubacterium]* was only found in the two groups treated with antibiotics. We also found that HEP can accessory inhibit all the above-mentioned bacteria, and the level of TNF-α was also inhibited.

Research has demonstrated that microbiota-activated PPAR-γ signaling is a homeostatic pathway that prevents the dysbiotic expansion of potentially pathogenic *Escherichia* and *Salmonella* by reducing the bioavailability of respiratory electron acceptors to *Enterobacteriaceae* in the lumen of the colon (Byndloss et al., 2017). In this study, there were some Unclassified_*Enterobacteriaceae*, *Enterobacter*, and *Proteus* in the antibiotics+5-Fu group, but the addition of HEP counteracted these side effects by inhibiting these pathobionts and increasing certain butyrate-producing bacteria including *Clostridium*, *Eubacterium*, and *Fusobacterium*. The PPAR signaling pathway was clearly activated as PPAR-α and PPAR-γ were increased in all the HEP+5-Fu groups (Figure 11 and 12), which indicated that HEP can help to counteract the side effects of antibiotics and thereby reduce the depletion of butyrate-producing microbes. PPAR-γ signaling is a key regulatory pathway for colorectal cancer (Milone et al., 2016; Panza et al., 2014; Portius, Sobolewski, & Foti, 2017), as PPAR-γ can regulate cell proliferation, differentiation, and survival, thereby controlling carcinogenesis in various tissues. Research has revealed that PPAR-γ can inhibit the expressions of Cyclin D1, NF-κB, c-MYC, P27, and VEGF (Portius et al., 2017), and we found the same results in this study. The levels of IFN-γ, IL-1β, IL-2, TNF-α, and IL-6 were inhibited, which may have been due to microbiota-activated PPAR-γ signaling.

In the gut, *Coriobacteriaceae* carry out important functions such as the conversion of bile salts and steroids as well as the activation of dietary polyphenols. However, they can also be considered as pathobionts, because their occurrence has been associated with a range of pathologies such as bacteremia, periodontitis, and vaginosis (Chen et al., 2012; Claus et al., 2011). In this study, the genus-level analysis showed that *Adlercreutzia* and Unclassified_*Coriobacteriaceae* were increased in the xenograft-only group ($P < 0.01$ compared with control), but HEP+5-Fu reduced the imbalances. Levels of other bacteria including *F16*, *Dorea*, *Prevotella*, *Desulfovibrionaceae*, *[Prevotella]*, *Anaeroplasma*, *Bilophila*, *Helicobacteraceae*, *Flexispira*, *Parabacteroides*, *Flavobacteriaceae*, *Streptococcus*, and *Christensenellaceae* were also changed. Data from *in vitro* and *in vivo* studies have indicated that consuming probiotic nonpathogenic lactic acid bacteria has beneficial effects on cancer (Ambalam, Raman, Purama, & Doble, 2016; Linn, Thu, & Win, 2019; Ranji, Akbarzadeh, & Rahmati-Yamchi, 2015; Redman, Ward, & Phillips, 2014; Wardill, Van Sebille, Ciorba, & Bowen, 2018). The abundance levels of some probiotic bacteria such as *Roseburia*, *Lactobacillus*, *Bifidobacterium*, and *Akkermansia* were increased, especially in the HEP+5-Fu groups. All these indicated that the HEP could enhance the auxiliary antitumor effects of 5-Fu by improving the gut microbiota.

Evidence is growing that the gut microbiota modulates the host response to chemotherapeutic drugs, with three main clinical outcomes: facilitation of drug efficacy, abrogation and compromise of anticancer effects, and mediation of toxicity (Louis, Hold, & Flin, 2014). The implication is that the gut microbiota is critical to the development of personalized cancer treatment strategies and, therefore, greater insight into prokaryotic cometabolism of chemotherapeutic drugs is now required. In our study, the KEGG pathway analysis of the gut microbiota metagenome were changed in functional genes coding for metabolism of xenobiotics by cytochrome P450, drug metabolism - cytochrome P450 and drug metabolism - other enzymes (Figure 10C). This means that the absorption and metabolism of the chemotherapeutic drug 5-Fu was influenced by the gut microbiota. Figure 12K shows that the serum concentrations of 5-Fu in HEP-treated groups were higher than those of corresponding 5-Fu-only groups (pre-xenograft HEP compared with pre-xenograft HEP+5-Fu; 5-Fu-only compared with HEP+5-Fu; antibiotics+HEP+5-Fucompared with antibiotics+5-Fu; 5-Fu-only compared with antibiotics+5-Fu), and HEP improved all the groups responses (Figure 10C). These results all indicated that HEP can enhance the absorption of 5-Fu may via improving the gut microbiota.

Fungal immunomodulatory proteins (FIPs) are small-molecule proteins extracted from the fruiting body of certain higher basidiomycetes (mushrooms) that are rapidly becoming recognized as a promising source of novel proteins. FIPs have a similar structure and immune functions as lectins and immunoglobulins, which have exhibited many beneficial functions in previous studies, including antitumor effects, anti-allergy effects, and immunity stimulation. FIPs have shown unique features and could offer solutions to several medical and biotechnological problems such as microbial drug resistance, low crop yields, and demands for renewable energy (Chen et al., 2017). In this study, we found that the gut microbiota influences the efficacy of 5-Fu, and HEP could improve the gut microbiota and thereby modulate immune inflammatory responses and promote the absorption of chemotherapeutic drugs. The metabolism and absorption of HEP (including the involvement of the gut microbiota) needs more research. The gut microbiota should now be targeted to improve the efficacy and reduce the toxicity of current chemotherapy agents. The rational use of probiotics, prebiotics, synbiotics, antibiotics, and diet could improve the efficacy and reduce the toxic burden of chemotherapeutic drugs, and surely gain increasing attention with regard to the field of host–bacteria interactions in the near future.

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CONSENT FOR PUBLICATION

No human data are reported in this manuscript.

AVAILABILITY OF DATA AND MATERIALS

The data sets supporting the conclusions of this paper are included within the paper and its additional files.

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AUTHOR CONTRIBUTIONS

All the authors participated in designing this study, carrying out the computational analyses and writing the manuscript. Wang and Zhu fed the mice, collected the physiological data and fecal samples and extracted microbial DNA. Wang and Tang help to collect the mice physiological data and brain samples. Wang, Chen, and Li collected data regarding the microbial metabolic networks and proteomic analysis. Tang and Zhu helped with the histopathological examination. Wang, Chen, Tang, and Zhu carried out the western blotting analysis. Chen and Xie helped to design the study and to develop the metagenomic analysis tools and reviewed the manuscript. All authors have read and approved the final manuscript.

ETHICS APPROVAL

The animal protocols used in this work were approved by the Institutional Animal Care and Use Committee of the Center of Laboratory Animals of the Guangdong Inst. of Microbiology.

COMPETING INTERESTS

All the authors declare that no competing interests exist.

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